

OSPF - Open Shortest Path First

Open Shortest Path First (OSPF) is a widely used routing protocol for Internet Protocol (IP) networks. It is a link-state routing protocol that falls under the category of Interior Gateway Protocols (IGPs), designed to work within a single autonomous system (AS).

Some Important Points about OSPF:

- OSPF Uses the Dijkstra Algorithm to find the best Path.
- It is Layer 3 protocol and the Protocol Number is 89.3
- OSPF is a Classless Routing Protocol
- OSPF Supports the Authentication also.
- OSPF Supports both IPV4 and IPV6

Some Important Points about OSPF:

- OSPF AD Value:
 - . Intra Area: 110
 - . Inter Area: 110
 - . External Area: 110
- OSPF Timers:
 - . Hello Timer: 10 Sec
 - . Dead Timer: 40 Sec
- OSPF Multicast Address:
 - . Hello: **224.0.0.5**
 - . Update: **224.0.0.6**

Routing Codes of OSPF:

- O: Single Area , OIA: Other Area
- O*E1, O*E2

Router Roles in OSPF:

- DR:** Designated Router
- BDR:** Backup Designated Router
- DR Other:** Designated Router other

(Note: The Election process followed by the OSPF DR and BDR on the basis highest Priority number and if Priority number is same then Highest Router ID will become the DR)
By Default OSPF Enable Priority is 1.

Process ID:

In OSPF (Open Shortest Path First), the **Process ID** is a unique identifier used to differentiate between multiple OSPF instances running on the same router. It allows a single router to participate in different OSPF routing domains simultaneously or to apply different OSPF configurations within the same router.

Configuration Example:

```
Router(config)#router ospf <process-id>
```

```
Router(config)#router ospf 100
```

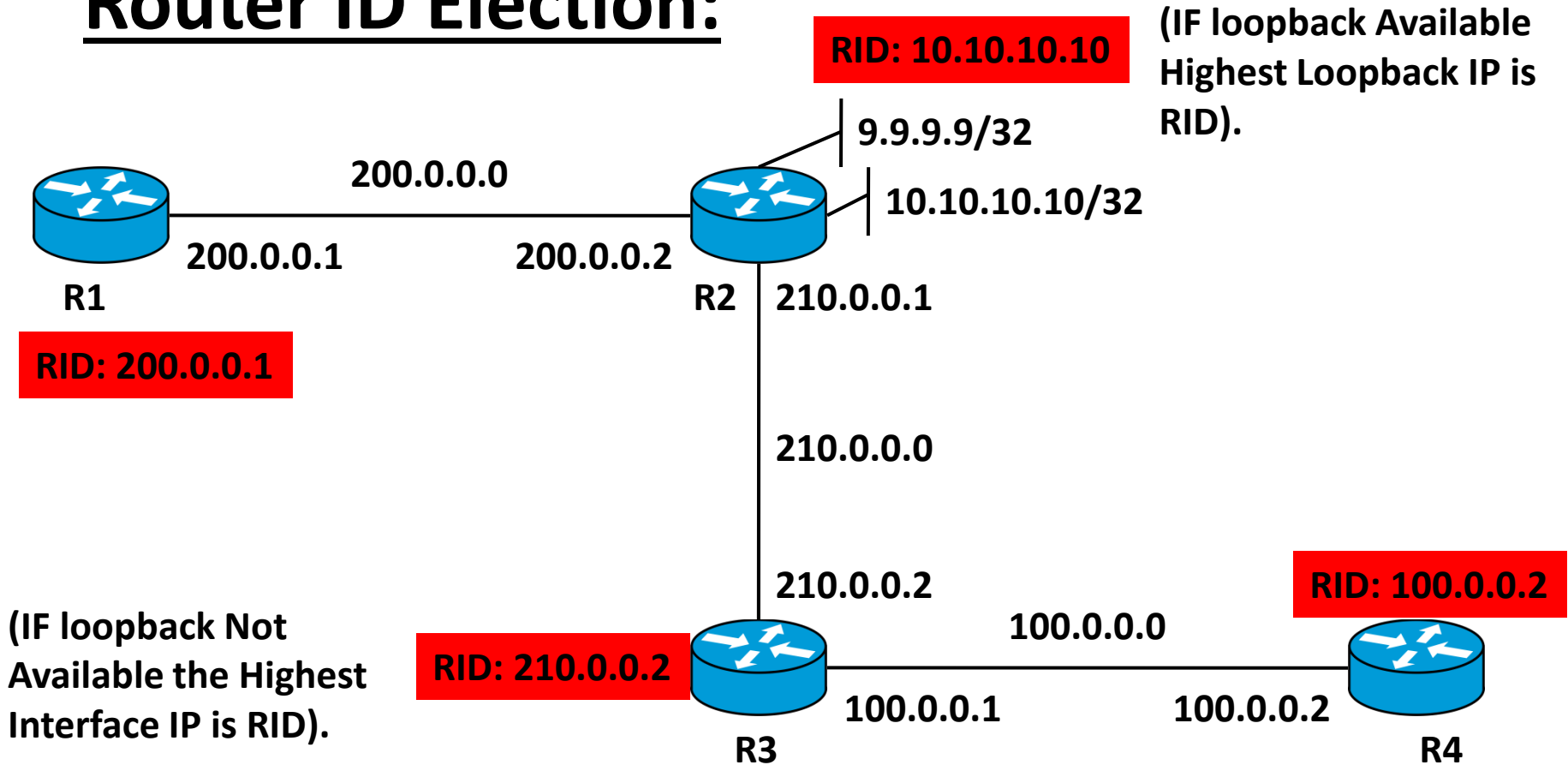
What is Router ID in OSPF?

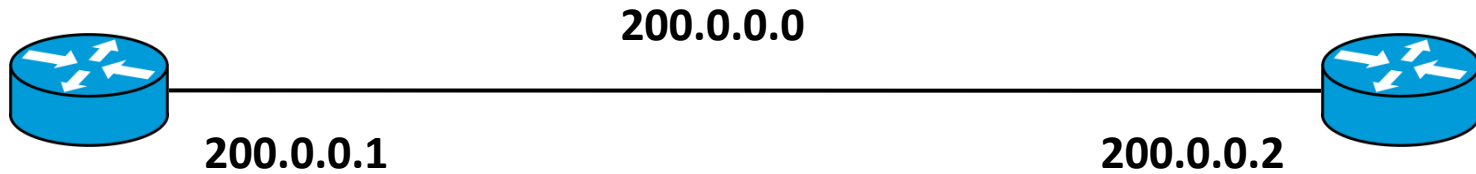
- It look Like IPV4 Address.
- Size of Router ID is 32 Bits.
- Every Device Router ID Should be unique.
- Router ID Helps to Create the DATABASE.

Router ID is elected by procedure:

- Manually Assign the Router ID.
- If loopback is available then highest Loopback Interface IP Will become Router ID.
- If Loopback IP is not available then Highest Physical Interface IP will be Router ID.

Router ID Election:





OSPF Configuration Example:

Router>en

Router#conf t

Router(config)#router ospf 1

Router(config-router)#network 200.0.0.0 0.0.0.255 area 0

Router(config-router)#ex